PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: LoanAI

Version: 1.0

1. Project Overview

- **Problem Statement:** The traditional loan application process within the Malaysian banking sector is characterized by significant manual effort, leading to inefficiencies, prolonged processing times, and a higher propensity for human error. The challenge is further compounded by the diverse and often unstructured nature of customer inputs, including free-form text and various document formats. This manual dependency results in delayed loan decisions, suboptimal customer experiences, and increased operational costs for financial institutions.
- **Target Domain:** This project targets the Malaysian Banking Sector, specifically focusing on the automation and enhancement of the loan application and processing workflow.
- **Proposed Solution Summary:** LoanAI is an intelligent, multi-agent loan processing system designed to revolutionize the loan application workflow for Malaysian banks. It employs an advanced AI-powered pipeline to seamlessly transform unstructured customer inputs—such as natural language text and uploaded documents (e.g., payslips, bank statements, identity cards)—into structured data, culminating in rapid and accurate loan decisions. The system orchestrates a series of specialized AI agents, including an Orchestrator, Extraction, Validation, Financial Analysis, Credit Check, and Decision Agent, to ensure a comprehensive and efficient process.

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2. Background & Business Objective

- **Background of the Problem:** The current landscape of loan processing in Malaysia, much like in many other regions, relies heavily on manual data entry, meticulous document verification, and subjective assessment by loan officers. This labor-intensive approach contributes to extended turnaround times, often spanning several days to weeks, which directly impacts customer satisfaction and escalates operational expenditures for banks. The escalating volume of loan applications, coupled with the market demand for expedited decision-making, underscores the critical need for an automated, intelligent solution capable of handling complex, unstructured data inputs efficiently.
- **Importance of Solving This Issue:**
  Addressing the inefficiencies in loan processing through LoanAI offers substantial benefits:
  - For Financial Institutions: It promises a significant reduction in operational costs, enhanced processing efficiency, accelerated loan disbursement cycles, improved data accuracy, more robust risk assessment capabilities, and strengthened compliance with regulatory standards. The automation frees up human resources to focus on more complex cases and customer relationship management.
  - For Customers: It translates into faster loan approvals, a more transparent application journey, and an overall superior customer experience, aligning with modern expectations for digital-first services.

- **Strategic Fit / Impact:** LoanAI is strategically aligned with the ongoing digital transformation initiatives prevalent within the Malaysian financial sector. By integrating cutting-edge AI technologies, particularly multimodal capabilities, it positions financial institutions to gain a competitive edge. The system directly addresses the imperative for increased efficiency and an improved customer experience within a highly regulated banking environment, fostering innovation and sustainable growth.

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3. Product Purpose

3.1. Main Goal of the System: The primary goal of LoanAI is to establish an AI-powered, end-to-end automated loan processing system. This system will efficiently convert diverse, unstructured customer inputs into structured data, enabling real-time, accurate loan decisions specifically tailored for the Malaysian banking sector.

3.2. Intended Users (Target Audience):

While the system's benefits extend to various stakeholders, the primary intended users are:

- **Bank Loan Officers/Underwriters**: To automate repetitive tasks, allowing them to concentrate on intricate cases, conduct deeper analysis, and make more informed, strategic decisions.
- **Bank Operations Staff**: To facilitate the efficient management, tracking, and auditability of loan applications, ensuring operational smoothness and compliance.
- **Customers (Indirectly)**: Although not direct users of the internal agent system, customers will significantly benefit from a faster, more transparent, and less cumbersome loan application experience.

4. System Functionalities

4.1. Description: LoanAI is conceived as an intelligent multi-agent system that comprehensively automates the entire loan application lifecycle, from the initial customer interaction to the final decision delivery. It is engineered to process both natural language queries and multimodal document uploads, intelligently orchestrating a suite of specialized AI agents. These agents collaboratively perform critical functions such as data extraction, information validation, sophisticated financial analysis, simulated credit checks, and ultimately, a definitive loan decision.

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4.2. Key Functionalities:

The core functionalities of LoanAI are designed to deliver a seamless, efficient, and transparent loan processing experience:

- Natural Language Chat Interface: Provides an intuitive and accessible channel for customers to interact with the system, enabling them to submit information, respond to queries, and receive updates in a conversational manner.
- Multimodal File Upload (JPG/PNG/PDF): Facilitates the automatic extraction of critical data from various document types, including payslips, bank statements, and identity cards (IC), leveraging the advanced capabilities of Gemini 2.0 Flash (Multimodal).
- Real-time Pipeline Visualization: Offers a transparent view of the loan application's progress, illustrating which agent is actively processing the application at any given moment. This enhances user trust and provides clarity on the workflow status.
- DSR Gauge and Interactive Result Cards: Presents the outcomes of financial analysis, including the Debt Service Ratio (DSR) and assessed risk level, through an interactive and easily understandable interface. This allows for quick comprehension of the decision factors.
- API Key Persistence via .env: Ensures robust security for server-side operations by managing and persisting API keys through environment variables, preventing hardcoding and unauthorized access.
- Comprehensive Agent Trailing: Implements a detailed logging mechanism that records all activities, decisions, and interactions of each AI agent throughout the processing pipeline. This feature is crucial for auditability, regulatory compliance, and post-mortem analysis.

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4.3. AI Model & Prompt Design

4.3.1. Model Selection: Gemini 2.0 Flash (Multimodal)

LoanAI utilizes Gemini 2.0 Flash as its core LLM. This model was selected for the following strategic reasons:

- Multimodal Native Intelligence: Its ability to process text, images, and PDFs natively is critical for extracting structured data from Malaysian NRICs, payslips, and bank statements without requiring separate OCR engines.
- Low Latency (Flash): The "Flash" variant provides the high-speed inference necessary for a real-time chat interface and a responsive multi-agent pipeline.
- Large Context Window: It handles long document uploads (e.g., 6-month bank statements) without losing context, ensuring all financial transactions are captured for DSR calculation.

4.3.2. Prompting Strategy: Multi-Step Agentic Prompting

The system employs a Multi-Step Agentic Prompting strategy. Rather than using a single prompt, the workflow is broken down into specialized tasks handled by distinct agents:

- Reasoning: This approach fits the use case because loan processing requires logical sequencing (Extraction → Validation → Analysis).
- Few-Shot Learning: For the Extraction Agent, we provide "Golden Examples" of Malaysian documents in the prompt to ensure the model understands local nuances like "EPF deductions" or "Allowance" vs. "Basic Salary."

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4.3.3. Context & Input Handling:

- Maximum Input Size: The system accepts up to 10MB per file and a total of 15 files per application.
- Oversized Inputs: If an input exceeds these limits or the model's token limit (e.g., a 200-page PDF), the system implements Intelligent Chunking. It prioritizes the first and last pages of bank statements (where summaries usually reside) and uses a "Map-Reduce" approach to summarize transaction history.
- Rejection Policy: Files that are completely unreadable or exceed 20MB are rejected with a graceful notification to the user via the chat interface.

4.3.4. Fallback & Failure Behavior:

To ensure "Fintech-grade" reliability, LoanAI includes the following safety nets:

- Self-Correction Loop: If the Decision Agent detects a hallucination (e.g., a DSR calculation that is mathematically impossible based on the extracted income), it triggers a Single Retry with a more restrictive "System Instruction" prompt.
- Validation Checkpoint: The Validation Agent acts as a filter. If the model returns off-topic or unusable responses, the system enters a Graceful Error State, asking the user to "Please re-upload a clearer image of your document."
- Human Escalation Path: For the hackathon prototype, any "Rejected" decision includes a "Request Manual Review" button, which simulates escalating the case to a human loan officer by logging the agent trail for audit.

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5. User Stories & Use Cases

- **User Stories:**
  - *As a loan applicant, I want to submit my loan application and supporting documents easily through a chat interface, so that I can get a quick decision.*
  - *As a bank loan officer, I want the system to automatically extract data from customer documents and perform initial checks, so that I can focus on complex cases and final approvals.*
  - *As a bank operations manager, I want to see the real-time status of all loan applications and an audit trail of each decision, so that I can ensure compliance and operational efficiency.*
- **Use Cases (Main Interactions):**

1. Loan Application Submission Flow:* User Action: Applicant initiates a loan application via the natural language chat interface.* System Response: Orchestrator Agent guides the applicant through required information and document uploads.* User Action: Applicant uploads payslips, bank statements, and IC (JPG/PNG/PDF).* System Response: Extraction Agent processes documents using Gemini 2.0 Flash, extracting relevant data.

2. Data Validation and Clarification Flow:* System Action: Validation Agent identifies any missing or unclear information.* System Response: Validation Agent prompts the applicant via chat for clarification or additional documents.* User Action: Applicant provides missing information or new documents.

3. Financial Analysis and Credit Check Flow:* System Action: Financial Analysis Agent calculates DSR and assesses risk level based on extracted data and bank policies.* System Action: Credit Check Agent simulates credit scoring (e.g., based on CCRIS/CTOS data) and property valuation (if applicable).

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4. Decision and Notification Flow:* System Action: Decision Agent synthesizes all information and provides a final verdict (Approved/Rejected/Conditional).* System Response: Decision Agent presents the decision, DSR Gauge, and interactive result cards to the applicant via the chat interface.* System Response: Loan officer is notified of the decision and can review the comprehensive agent trailing for audit.

6. Features Included (Scope Definition)

- Multi-Agent Architecture: Orchestrator, Extraction, Validation, Financial Analysis, Credit Check, and Decision Agents.
- Natural Language Processing (NLP) Interface: For conversational interaction with applicants.
- Multimodal Data Extraction: Support for JPG, PNG, PDF document uploads using Gemini 2.0 Flash.
- Automated DSR Calculation: Based on Malaysian banking standards.
- Risk Level Assessment: Algorithmic determination of applicant risk.
- Simulated Credit Scoring: Integration with simulated credit bureau data (e.g., CTOS/CCRIS principles).
- Real-time Pipeline Visualization: Graphical representation of agent workflow status.
- Interactive DSR Gauge and Result Cards: User-friendly display of loan decision and key metrics.
- Secure API Key Management: Via .env files for server-side security.
- Comprehensive Agent Trailing: Detailed logging for auditability and transparency.
- Basic Property Valuation Simulation: For relevant loan types.

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7. Features Not Included (Scope Control)

To maintain focus for the hackathon and ensure timely delivery, the following features are explicitly out of scope for Version 1.0:

- **Direct Integration with Live Banking Systems**: No direct API integration with actual core banking systems, credit bureaus (CTOS, CCRIS), or property valuation databases. All credit checks and valuations will be simulated.
- **Advanced Fraud Detection Modules**: While risk assessment is included, sophisticated fraud detection mechanisms are not part of this initial scope.
- **Loan Disbursement and Servicing**: The system focuses solely on the decision-making process, not the post-approval stages of loan disbursement, repayment tracking, or customer service.
- **Complex Loan Product Support**: Initial focus will be on a single, defined loan product (e.g., personal loan or basic housing loan), not a wide array of complex financial products.
- **Multi-language Support**: The chat interface will initially support English only.
- **User Authentication and Authorization for Applicants**: The focus is on the agent pipeline; applicant login/account management is not included.

8. Assumptions & Constraints

- Availability of Gemini 2.0 Flash API: Assumes stable and accessible API for multimodal data extraction.
- Simulated Data Accuracy: Assumes that simulated credit scores and property valuations are sufficiently representative for hackathon demonstration purposes.
- Defined Loan Product Parameters: Assumes clear and predefined criteria for the loan product being processed.
- Input Document Quality: Assumes that uploaded documents are legible and in standard formats (JPG, PNG, PDF) for effective OCR and data extraction.
- Malaysian Regulatory Framework: Assumes a foundational understanding of Malaysian banking regulations for DSR and credit checks, even if direct integration is out of scope.

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9. Risks & Questions Throughout Development

- Hackathon Time Limit: Development is constrained by the hackathon timeline, necessitating a focused scope.

- Resource Limitations: Limited development resources (team size, infrastructure) typical of a hackathon project.

- No Direct PII Handling: Due to the hackathon environment and security considerations, the system will not handle actual Personally Identifiable Information (PII) for real customers. Simulated data will be used.

- No Production Deployment: The system is a prototype for demonstration purposes and not intended for production deployment without further development and security audits.

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